

TITANIC DATASET

Exploratory Data Analysis

DATA ANALYST INTERNSHIP — TASK 5

PROJECT OBJECTIVE

Explore passenger data using statistical analysis and visualizations to identify trends and relationships.

TOOLS USED

Python, Pandas, Matplotlib, Seaborn

DATASET OVERVIEW

Passenger records: 891

Columns after cleaning: 11

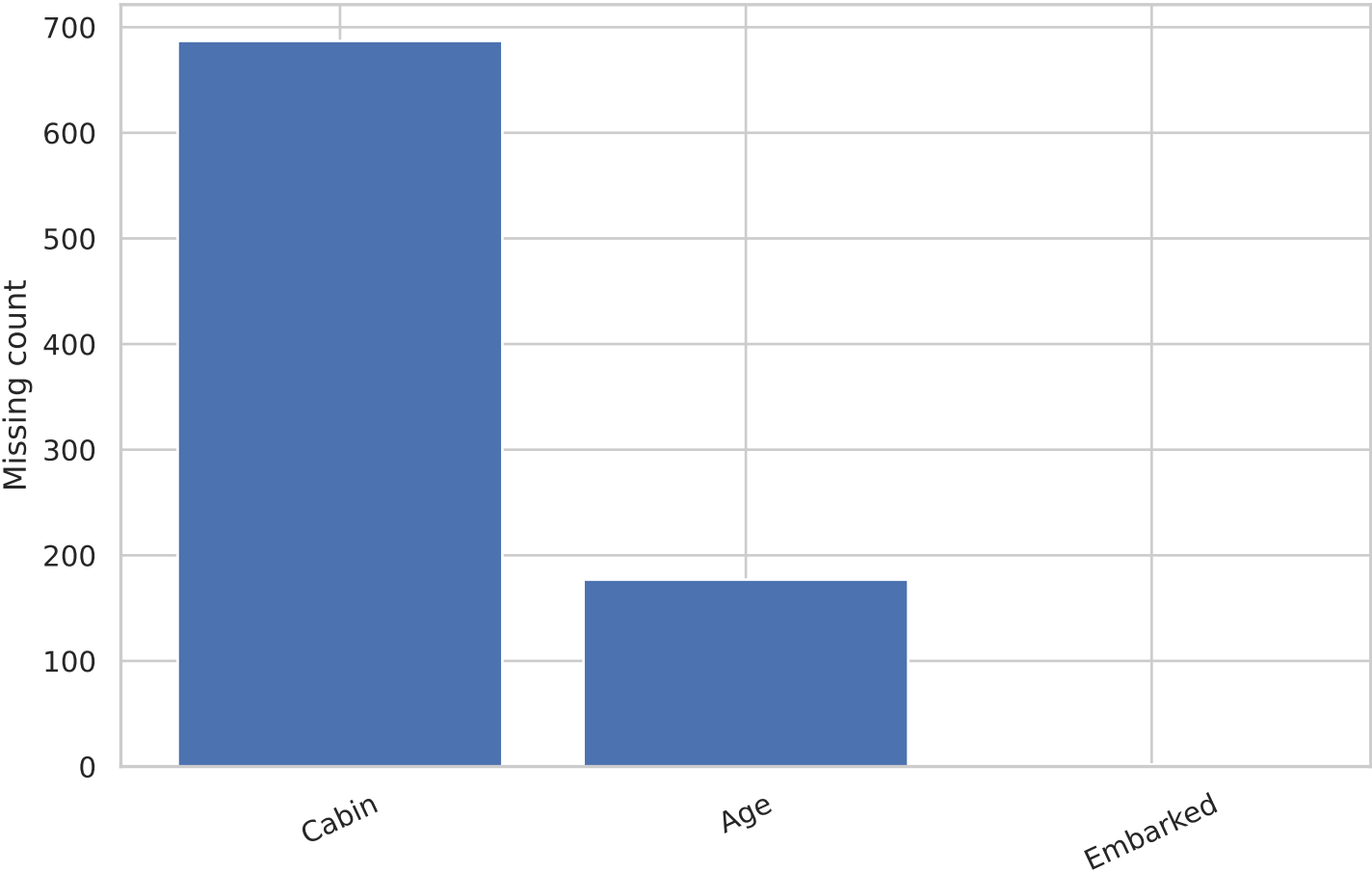
Overall survival rate: 38.38%

ANALYSIS PERFORMED

- Data inspection and missing-value analysis
- Descriptive statistics and frequency counts
- Survival analysis by gender and passenger class
- Histograms, boxplots and scatterplots
- Correlation heatmap and pairplot

Dataset Inspection and Statistics

Missing Values in Original Dataset

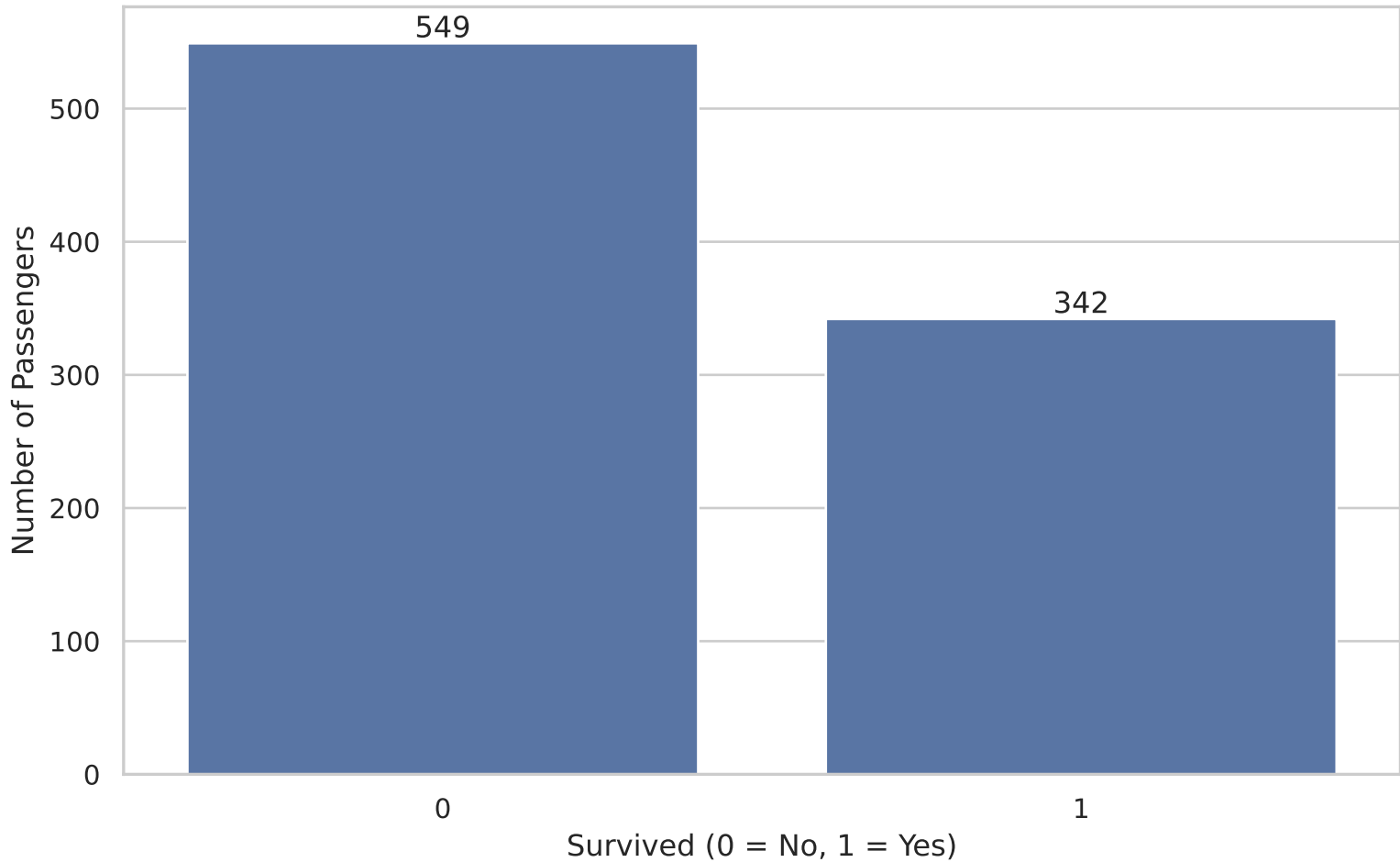


Descriptive Statistics

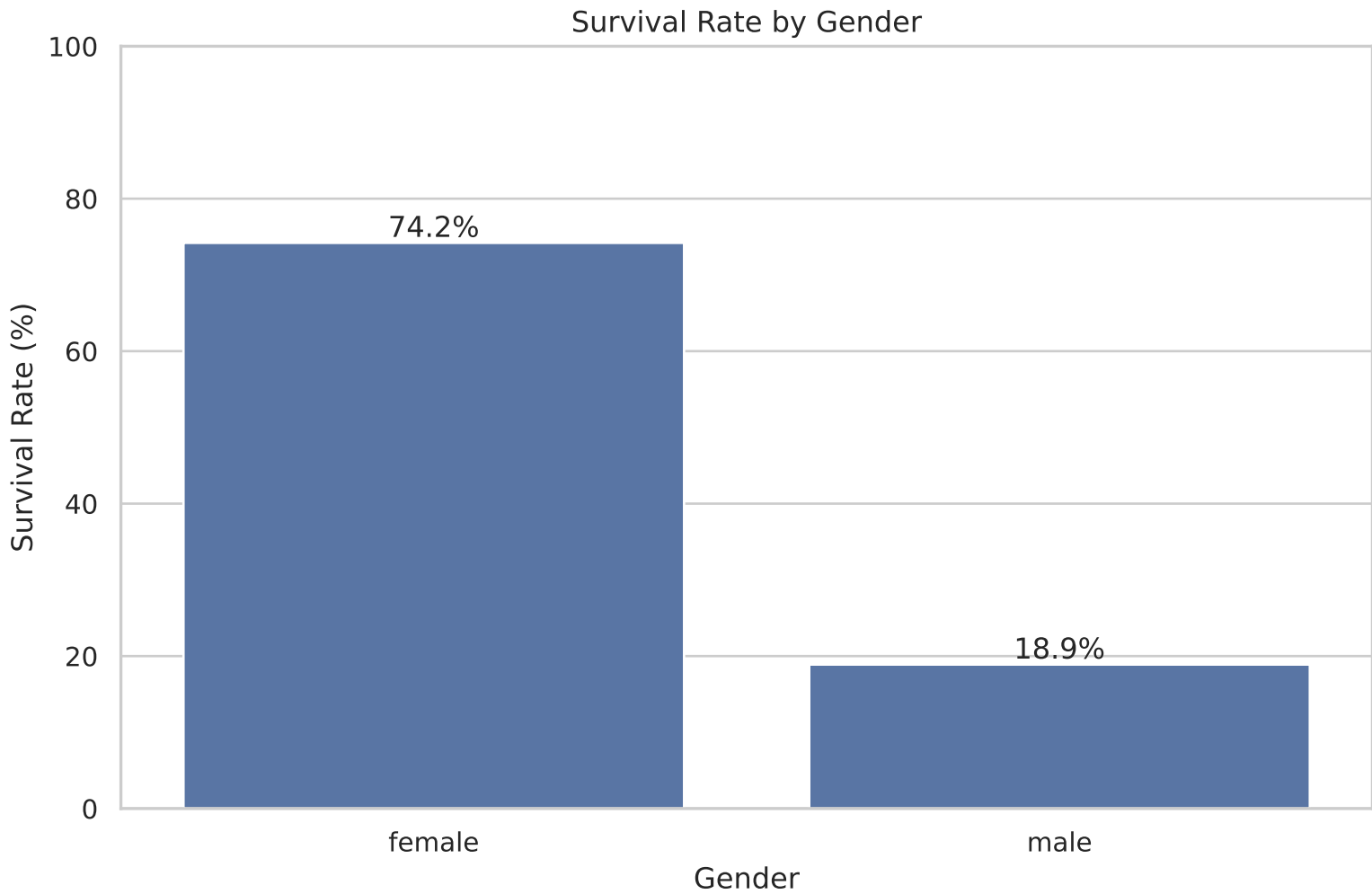
	PassengerId	Survived	Pclass	Age	SibSp	Parch	Fare
count	891.0	891.0	891.0	891.0	891.0	891.0	891.0
mean	446.0	0.38	2.31	29.36	0.52	0.38	32.2
std	257.35	0.49	0.84	13.02	1.1	0.81	49.69
min	1.0	0.0	1.0	0.42	0.0	0.0	0.0
25%	223.5	0.0	2.0	22.0	0.0	0.0	7.91
50%	446.0	0.0	3.0	28.0	0.0	0.0	14.45
75%	668.5	1.0	3.0	35.0	1.0	0.0	31.0
max	891.0	1.0	3.0	80.0	8.0	6.0	512.33

Cleaning: Age filled with median; Embarked filled with mode; Cabin excluded due to extensive missing data.

Passenger Survival Distribution

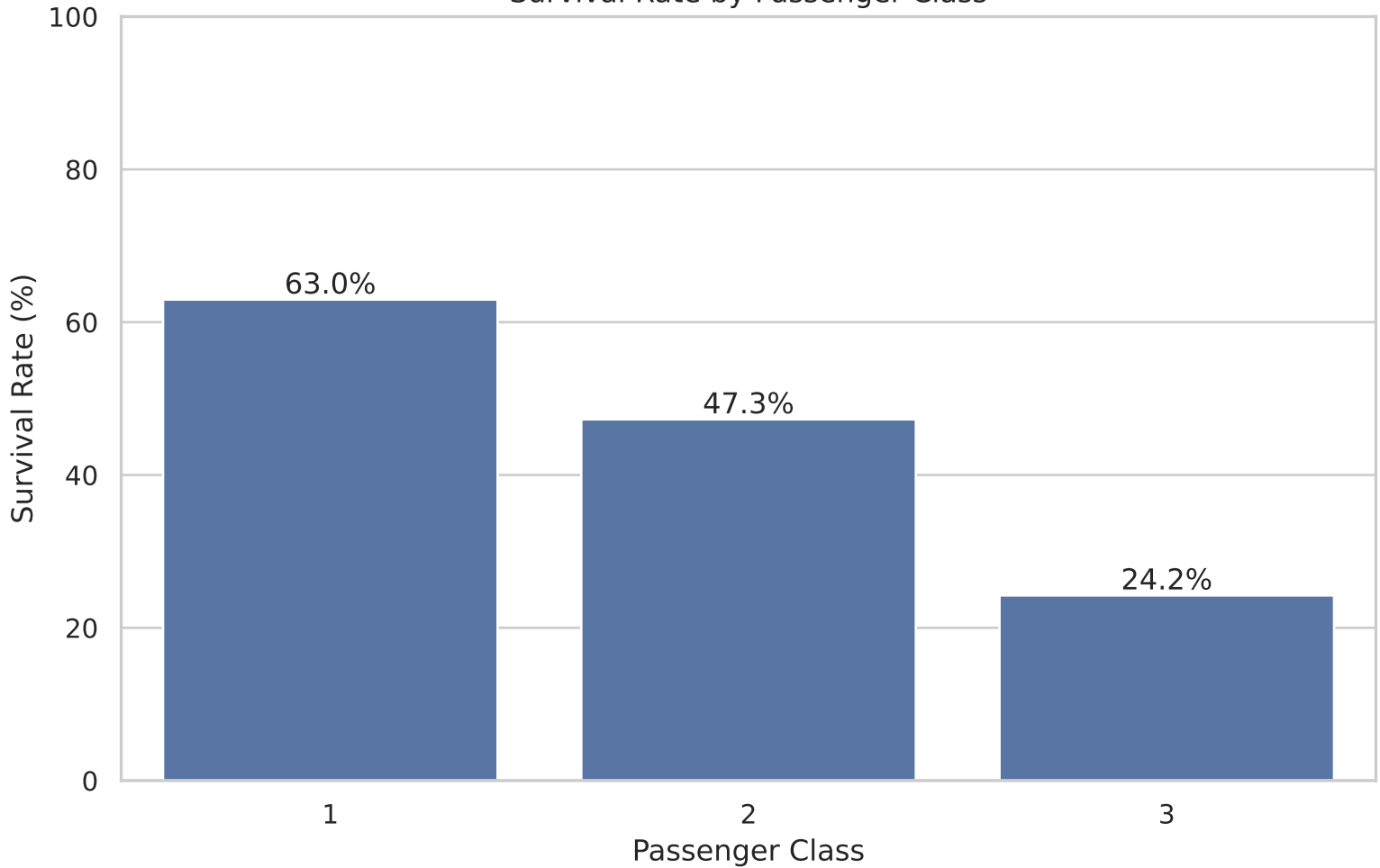


Observation: 342 passengers survived and 549 did not. The overall survival rate was 38.38%.



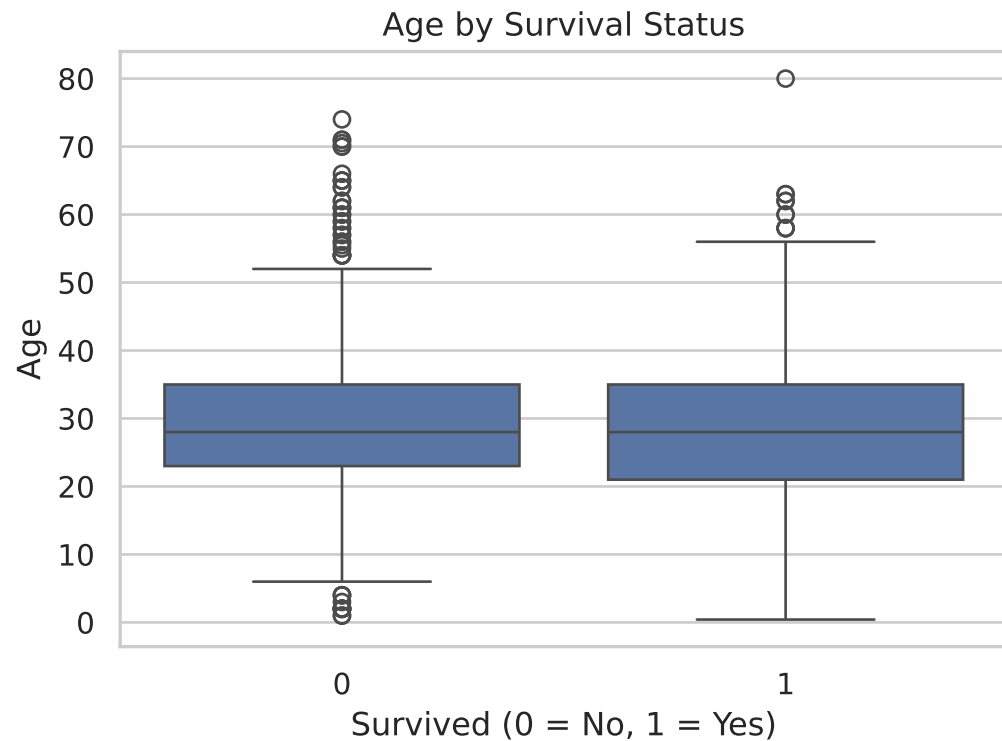
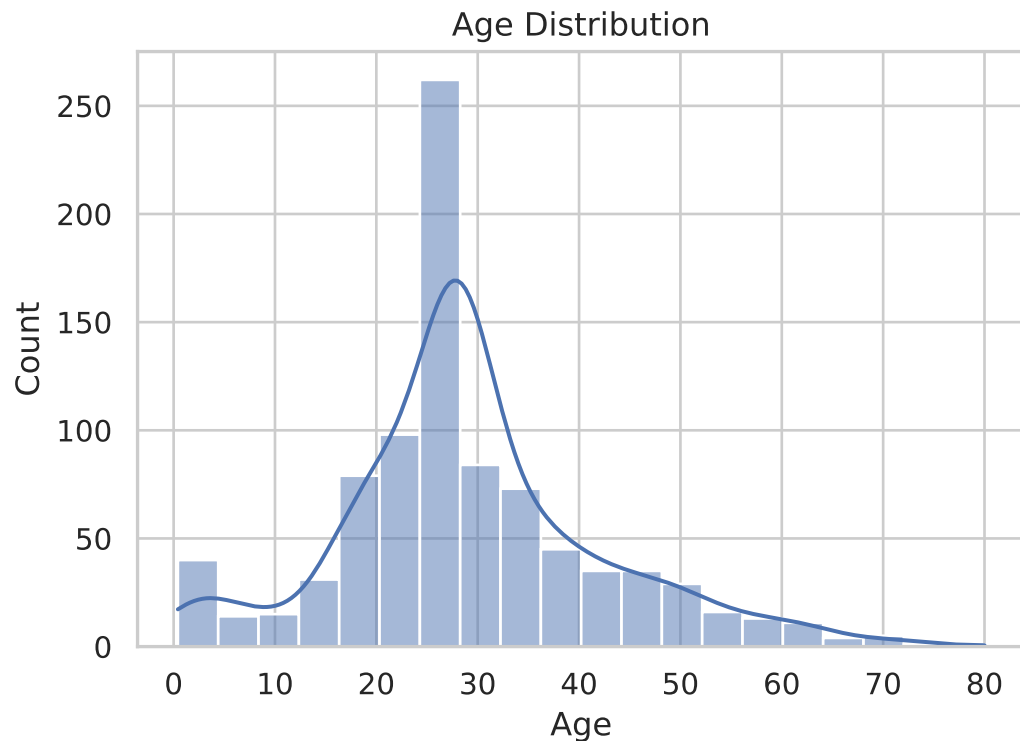
Observation: Compare the survival percentages for female and male passengers to identify gender-related differences.

Survival Rate by Passenger Class



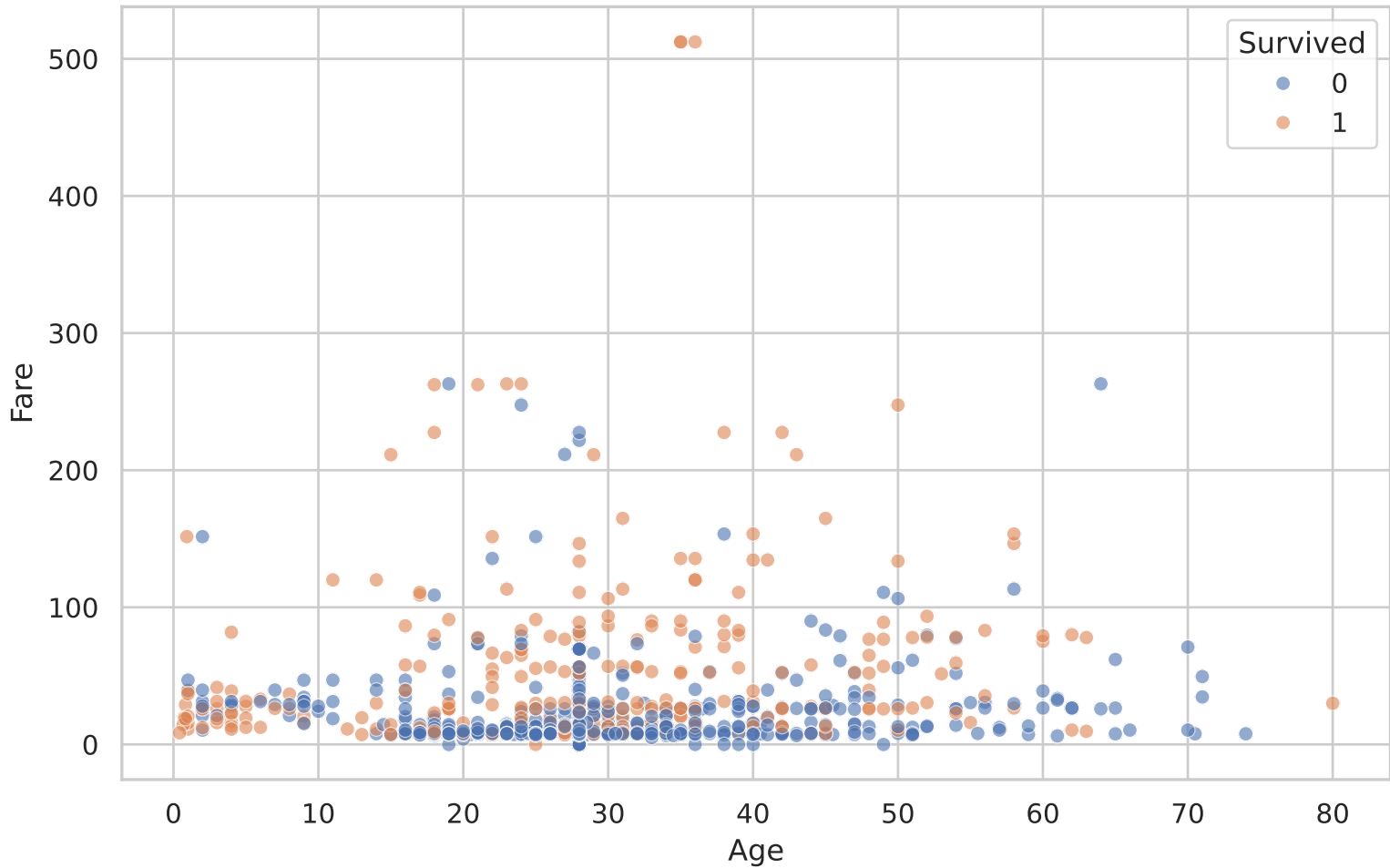
Observation: Compare survival rates across first, second, and third class passengers.

Age Analysis

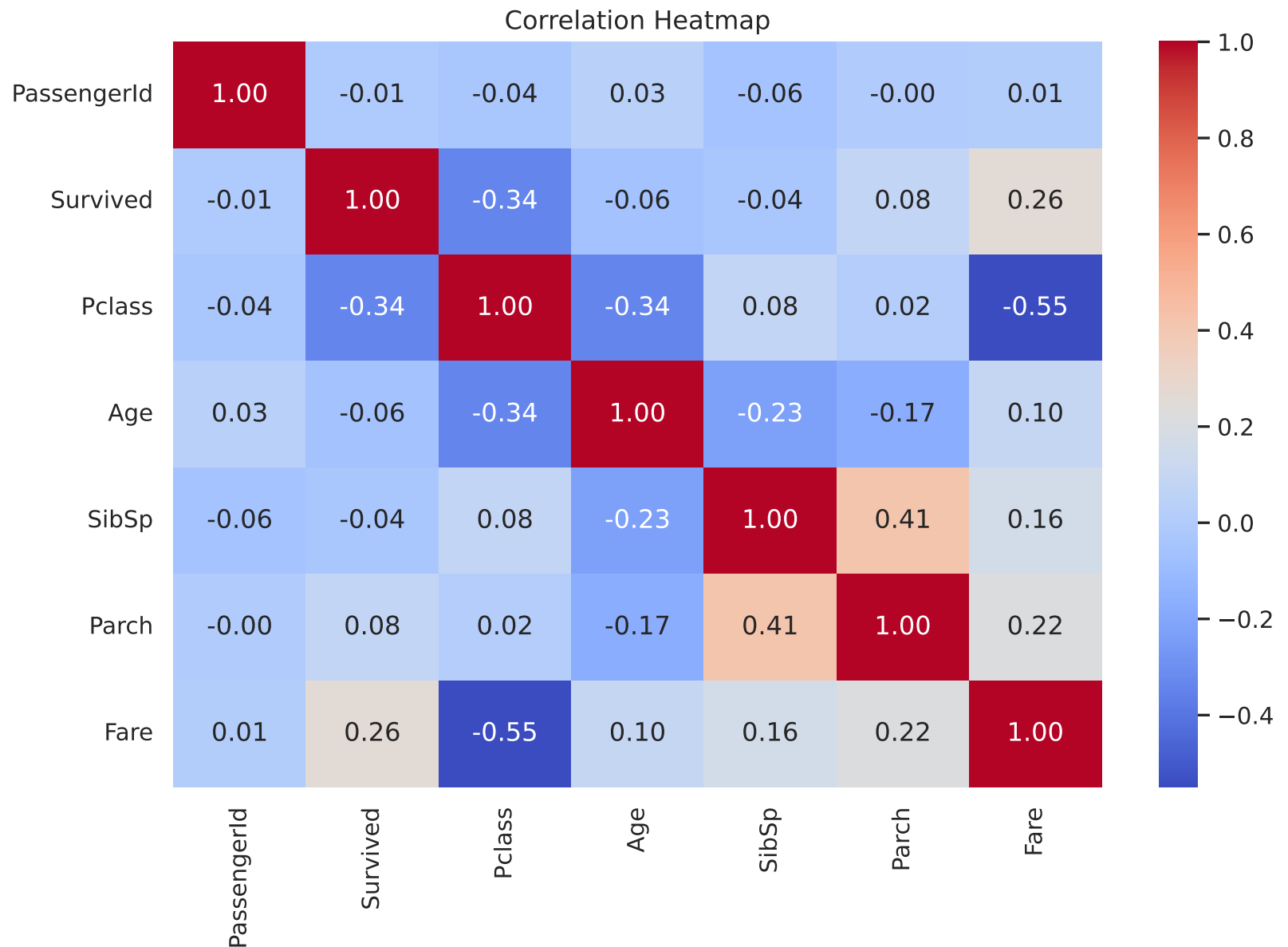


Observation: The histogram shows age distribution. The boxplot compares medians, spread, and potential outliers between survival groups. Missing ages were median-imputed.

Passenger Age vs Fare by Survival

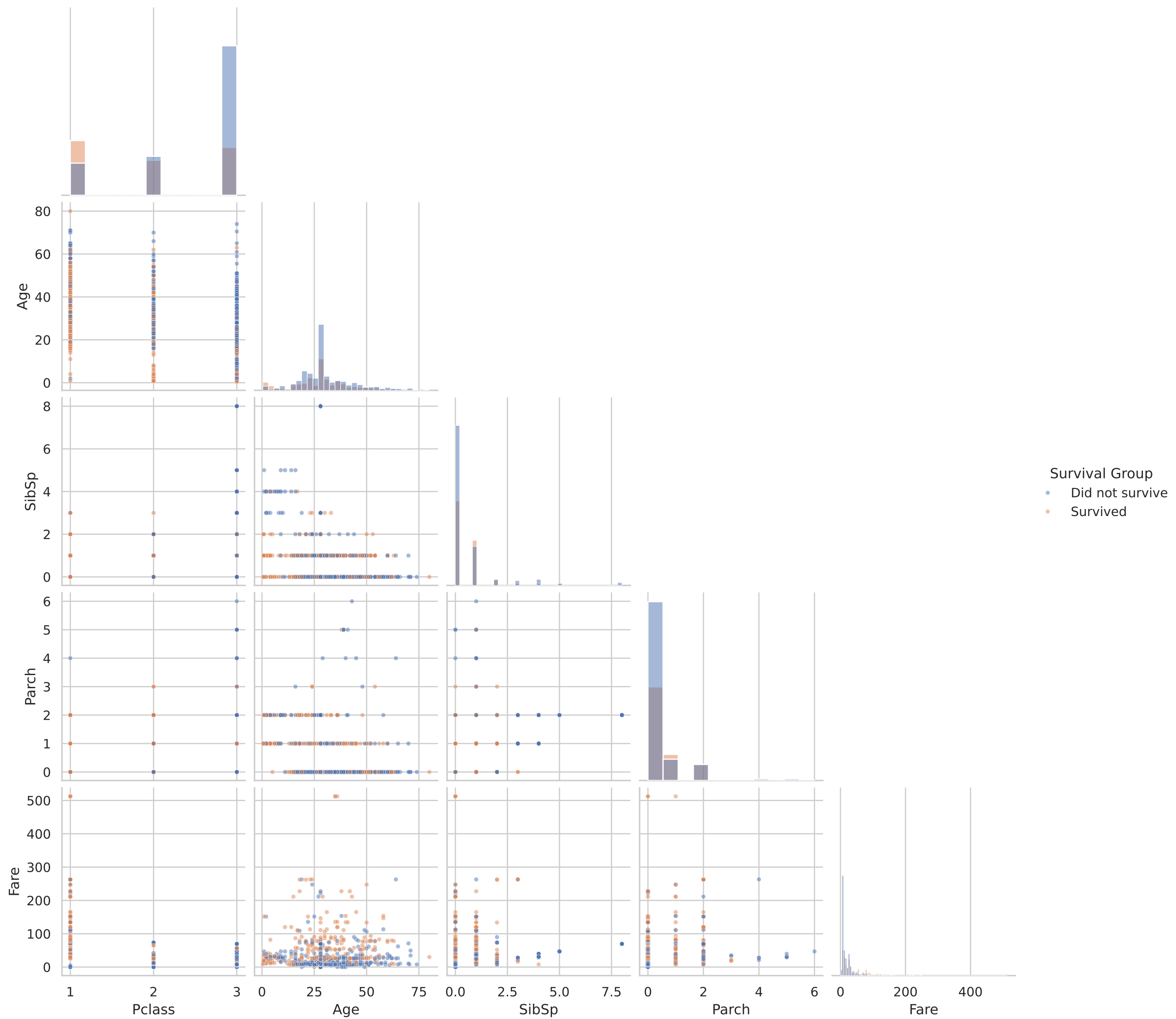


Observation: Each point represents a passenger. Look for visible patterns between age, fare, and survival.



Observation: Values closer to +1 or -1 indicate stronger linear relationships. Correlation does not imply causation.

Pairplot: Relationships Between Passenger Variables



Key Findings and Conclusion

Overall survival rate: 38.38%.

Female survival rate: 74.20%.

Male survival rate: 18.89%.

First-class survival rate: 62.96%.

Second-class survival rate: 47.28%.

Third-class survival rate: 24.24%.

CONCLUSION

The analysis used statistical summaries and visualizations to investigate passenger survival patterns and relationships among variables in the Titanic dataset.

DATA LIMITATIONS

Missing ages were filled using the median, and the Cabin column was excluded because of extensive missing data. These decisions may affect some interpretations. Correlation does not establish causation.